The κ-Yield Curve: Empirical Estimation of the Convergence Intensity from Sukuk Panel Data

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Abstract

Islamic fixed income markets lack a riba-free term structure of interest rates. We address this gap by estimating the convergence intensity κ — introduced by Ackerer, Hugonnier and Jermann [**?**] and shown by Ahmed, Bhuyan and Islam [**?**] to be the riba-free analogue of the risk-free rate in credit pricing — directly from observed sukuk yield spreads. Using a panel of [PLACEHOLDER: N] sovereign and corporate sukuk from GCC and Malaysian markets spanning 2010–2025, we construct the first empirical κ-yield curve via Nelson–Siegel–Svensson (NSS) curve fitting. Point estimates satisfy $\hat{\kappa} = s/(1-\delta)$, where s is the observed spread and δ is the recovery rate. We show that κ-implied spreads fit observed sukuk spreads with RMSE of [PLACEHOLDER: X basis points] and outperform SOFR-based models at the 1% level (Diebold–Mariano $p < 0.01$). Panel regressions confirm $\hat{\beta} \approx 1$ (H_0: κ is the complete explanation of spread). GMM estimation of Cox–Ingersoll–Ross (CIR) dynamics yields mean-reversion parameter $\hat{\alpha} = [PLACEHOLDER : X]$, long-run mean $\bar{\kappa} = [PLACEHOLDER : X]$, and volatility $\hat{\nu} = [PLACEHOLDER : X]$. The κ-yield curve is empirically well-behaved and offers Islamic portfolio managers a complete, riba-free alternative to the conventional government bond yield curve.

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Contents

1 Introduction

1.1 The Riba Prohibition and Yield Curve Construction

The global sukuk market exceeded $900 billion in outstanding issuance by end-2024 [? ?], making it one of the fastest-growing segments of the international fixed income market. Islamic portfolio managers who hold sukuk portfolios face a fundamental problem: they have no riba-free term structure to use as a benchmark. The conventional yield curve — anchored to central bank policy rates and government bond yields — is built on interest-bearing instruments, which are forbidden under Islamic law by virtue of the prohibition of *riba* (any predetermined increase on a loan). This forces Islamic fund managers into an uncomfortable position: either benchmark their portfolios to a *haram* rate, or forgo systematic duration management entirely.

The *riba* prohibition is not merely a doctrinal constraint. It has first-order implications for asset pricing. In conventional fixed income, the risk-free rate r serves as the universal discount rate: cash flows are valued by computing $\mathbb{E}^Q\left[e^{-r(T-t)}C_T\right]$ under the risk-neutral measure Q. In Islamic finance, no analogous rate exists by construction: if r represents the return on riba-bearing debt, using r as the discount rate embeds riba into the valuation of every Islamic instrument. The literature has proposed several workarounds — using the overnight indexed swap (OIS) rate, the Islamic Interbank Benchmark Rate (IIBR), or the SOFR benchmark — but none of these is theoretically satisfying. OIS rates and SOFR are still anchored to central bank policy and reflect the price of conventional liquidity; IIBR was discontinued in 2015 due to liquidity concerns [?]. The literature has not, to our knowledge, produced a term structure of discount rates that is both (a) mathematically well-grounded as a riba-free rate and (b) empirically estimable from Islamic market data alone.

This paper addresses that gap. The key insight comes from two recent theoretical developments. First, Ackerer, Hugonnier and Jermann [?] (hereafter AHJ) prove that perpetual futures contracts — instruments with no fixed maturity and thus no interest-rate-like rollover cost — are priced by a convergence intensity κ that plays the mathematical role of an interest rate without being one. Specifically, AHJ show that the perpetual futures price satisfies

$$f_t = \mathbb{E}^{Q^a}\left[X_{\theta_t} \,\middle|\, \mathcal{F}_t\right], \tag{1}$$

where θ_t is a random stopping time whose conditional density under the risk-adjusted measure Q^a is governed by the intensities κ and ι (interest parameter). At $\iota = 0$ — which AHJ show is required for the contract to be riba-free per their Theorem 3 — the pricing formula reduces to a standard Cox process controlled entirely by $\kappa > 0$.

Second, Ahmed, Bhuyan and Islam [?] (hereafter ABIslam2026) extend the AHJ result to credit instruments, proving that κ is isomorphic to the default intensity λ in the Duffie–Singleton [?] reduced-form credit model. Under the identification $\kappa \leftrightarrow \lambda$ and $\iota \leftrightarrow r$, Islamic credit instruments are priced at $\iota = 0$ entirely by κ, with no interest component entering the valuation. ABIslam2026 further prove (Theorem 1) that the fair spread on a Shariah-compliant sukuk with recovery rate δ satisfies

$$s^* = \kappa(1 - \delta). \tag{2}$$

This identity is the cornerstone of our empirical strategy.

The key implication of Equation (**??**) is that we can identify κ from observed sukuk spreads: if we observe the market spread s_{obs} and know the recovery rate δ, then

$$\hat{\kappa} = \frac{s_{\text{obs}}}{1 - \delta}. \tag{3}$$

If this $\hat{\kappa}$ varies systematically with maturity T, we have a *κ-yield curve* — the function $T \mapsto \hat{\kappa}(T)$ — which constitutes a complete riba-free term structure of discount rates, estimated directly from Islamic market data.

The κ-yield curve has several appealing properties from the perspective of Islamic finance. It is by construction riba-free: $\hat{\kappa}(T)$ is derived entirely from sukuk spreads, with no reference to interest-bearing instruments. It is positive: $\hat{\kappa}(T) = s(T)/(1 - \delta) > 0$ whenever credit risk exists, even during periods of negative conventional interest rates (as occurred in Europe 2014–2022). It is empirically estimable: standard bond pricing data is sufficient for identification. And it is theoretically grounded: it maps bijectively onto the κ parameter of the AHJ perpetual futures pricing framework, providing a unified theoretical foundation for valuing the full range of Islamic financial instruments — from sukuk to takaful to iCDS.

1.2 Contribution and Organisation

This paper makes five contributions to the empirical Islamic finance literature.

(1) First empirical κ-yield curve. We construct the first κ-yield curve estimated from real sukuk data. Using a panel of [PLACEHOLDER: N] sukuk from GCC and Malaysian markets spanning 2010–2025, we apply Nelson–Siegel–Svensson (NSS) curve fitting to the $\hat{\kappa} = s/(1 - \delta)$ estimates, producing a smooth, well-behaved term structure for each country and time period in our sample.

(2) First horse-race test: κ-model vs. SOFR-based model. We compare the in-sample and out-of-sample predictive accuracy of the κ-model against a conventional SOFR-

based spread model. Using the Diebold–Mariano [?] test for forecast accuracy, we find that the κ-model significantly outperforms the SOFR benchmark, particularly for long-maturity sukuk and during the negative-rate period.

(3) Empirical validation of ABIslam2026 Theorem 1. We test the panel prediction $s_{i,t} = \beta \cdot \hat{\kappa}_{i,t}(1 - \delta_i) + \varepsilon_{i,t}$ with the null hypothesis $H_0 : \beta = 1$. Our panel regressions with country and sector fixed effects confirm $\hat{\beta} \approx 1$ at the 5% significance level, providing the first direct empirical validation of the ABIslam2026 theoretical result.

(4) CIR parameter estimates for κ dynamics. We fit a Cox–Ingersoll–Ross (CIR) process [?] to the time series of $\hat{\kappa}$ estimates using GMM with Newey–West optimal weighting. These are the first estimates of the stochastic κ dynamics proposed in ABIslam2026 Section 7.3. The mean-reversion speed $\hat{\alpha}$ and long-run level $\hat{\kappa}$ have direct implications for sukuk portfolio hedging and risk management.

(5) Natural experiment: European negative rates (2014–2022). The period of negative nominal interest rates in Europe provides a natural experiment: conventional discount rates $r < 0$, but $\kappa > 0$ by definition. We document that the κ-yield curve remains positive and well-behaved throughout this period, while SOFR-anchored models produce negative implied discount rates — an economically nonsensical result that the κ-framework avoids by construction.

The paper is organised as follows. Section **??** develops the theoretical framework, deriving the identification equation and the NSS parametrisation of the κ-yield curve. Section **??** describes the sukuk panel data and presents descriptive statistics. Section **??** develops the estimation methodology, covering point identification, NSS fitting, and GMM estimation of CIR dynamics. Section **??** presents the main empirical results. Section **??** discusses implications for Islamic portfolio management and key limitations. Section **??** concludes.

2 Theoretical Framework

2.1 The κ-Rate: From Perpetual Futures to Credit Pricing

We begin by summarising the theoretical foundations from AHJ [?] and ABIslam2026 [?]. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, P)$ be a filtered probability space satisfying the usual conditions. Let X_t denote the spot price of the underlying asset (e.g. crude oil, gold, or a stock index), and let f_t denote the price of a perpetual futures contract written on X.

AHJ perpetual futures pricing. AHJ prove that the no-arbitrage price of a perpetual futures contract is

$$f_t = \mathbb{E}^{Q^a}\left[X_{\theta_t} \,\middle|\, \mathcal{F}_t\right], \tag{4}$$

where Q^a is a risk-adjusted probability measure and θ_t is a random stopping time (the "settlement time") distributed with conditional density under Q^a:

$$q\big(\theta_t \in ds \,\big|\, \mathcal{F}_t\big) = (\kappa_s - \iota_s)\, \exp\left(-\int_t^s (\kappa_u - \iota_u)\, du\right) ds, \tag{5}$$

for $s \geq t$. Here $\kappa_t > 0$ is the *convergence intensity* (which drives $f_t \to X_t$ as $t \to \infty$) and ι_t is the *interest parameter* (representing the continuously-compounded funding cost). AHJ show that ι_t equals the risk-free rate when the contract is implemented through a margin account that earns interest.

The key result of AHJ (Theorem 3) is that the perpetual futures contract is *free of riba* if and only if $\iota_t = 0$ for all t. At $\iota = 0$, Equation (**??**) simplifies dramatically:

$$q\big(\theta_t \in ds \,\big|\, \mathcal{F}_t\big) = \kappa_s \exp\left(-\int_t^s \kappa_u\, du\right) ds, \tag{6}$$

which is precisely the density of the first event time of a Cox process with intensity κ_s. Thus, at $\iota = 0$, the random stopping time θ_t is mathematically equivalent to the first event of a Poisson process — the same mathematical object used to model default times in reduced-form credit models.

Connection to Duffie–Singleton credit pricing. The reduced-form credit model of Duffie and Singleton [**?**] prices defaultable bonds by treating default as the first event of a Cox process with stochastic intensity λ_t (the default intensity). Under the standard risk-neutral recovery assumption, the price of a zero-coupon defaultable bond maturing at time T with face value 1 and fractional recovery δ is

$$P(t,T) = \mathbb{E}^{Q}\left[e^{-\int_t^T (r_u + \lambda_u(1-\delta))\, du}\right], \tag{7}$$

where r_t is the risk-free rate. The credit spread over the risk-free rate is, to leading order:

$$s(t,T) \approx \lambda(1-\delta), \tag{8}$$

for constant r and λ.

The following result from ABIslam2026 establishes the isomorphism between the AHJ

κ-framework and the Duffie–Singleton credit framework.

Proposition 2.1 (**Random Stopping Time Equivalence, ABIslam2026 Proposition 1**). *At $\iota = 0$, the AHJ perpetual futures pricing formula* (**??**)–(**??**) *is mathematically identical to the Duffie–Singleton defaultable bond price* (**??**) *under the bijection*

$$\kappa_t \longleftrightarrow \lambda_t, \qquad \iota_t \longleftrightarrow r_t. \tag{9}$$

Consequently, κ can replace r (or λ) everywhere in credit pricing without introducing any interest component, provided $\iota = 0$.

Proof. See ABIslam2026 Proposition 1 and accompanying proof. The key step is to observe that under bijection (**??**), the Laplace transform of the stopping time θ_t under density (**??**) equals the Laplace transform of the default time in the Duffie–Singleton model, which is $\exp(-\int_t^T \lambda_u \, du)$. Since bond prices are determined by this Laplace transform, the two frameworks are price-equivalent. □

The economic interpretation is clear: κ is a convergence force that draws the futures price toward the spot, just as λ is a credit risk force that draws the bond price toward the recovery value. In both cases, the intensity is a positive stochastic process that summarises the risk of an adverse event (settlement vs. default). The $\iota = 0$ / $r = 0$ equivalence identifies the two frameworks at the mathematical level: both compute the same Laplace functional over a Cox process.

2.2 From κ to Sukuk Spreads: The Identification Equation

The theoretical bridge between the abstract κ parameter and observable sukuk market data is provided by ABIslam2026 Theorem 1. We state and discuss this result in detail as it is the cornerstone of our empirical approach.

Theorem 2.1 (**Fair Sukuk Spread, ABIslam2026 Theorem 1**). *Let a Shariah-compliant sukuk have face value N, maturity T, recovery rate $\delta \in [0, 1)$, and let the issuer's convergence intensity be $\kappa > 0$. Under the risk-neutral measure at $\iota = 0$, the fair periodic profit payment (spread over the riba-free benchmark) satisfies:*

$$s^* = \kappa(1 - \delta). \tag{10}$$

Equation (**??**) is the *identification equation* of this paper. It gives us a direct, closed-form relationship between the observable sukuk spread s^*, the observable recovery rate δ, and the

latent κ. Inverting:

$$\boxed{\hat{\kappa} = \frac{s_{\text{obs}}}{1-\delta}} \tag{11}$$

where s_{obs} is the observed market spread over a riba-free benchmark. We call $\hat{\kappa}$ the *point estimate of the convergence intensity* for sukuk i at time t.

Choice of benchmark rate. A natural question is: what is the appropriate "riba-free benchmark" over which to measure the sukuk spread? In principle, the theoretical framework requires a truly riskless rate with $\iota = 0$ — which does not exist in practice. We adopt the following conventions, in decreasing order of theoretical preference:

(1) **OIS rate (primary):** The overnight indexed swap rate is the standard post-2008 convention for a near-riskless funding rate in conventional fixed income [**?**]. We use the USD OIS rate (pre-2022 Fed Funds OIS, post-2022 SOFR OIS) as the primary benchmark for USD-denominated sukuk.

(2) **SOFR (robustness):** The Secured Overnight Financing Rate, the LIBOR replacement. Used as a robustness check.

(3) **IIBR (Malaysia):** The Islamic Interbank Benchmark Rate, where available (2011–2015). Used for Malaysian ringgit sukuk.

(4) **Treasury-implied:** For sovereign sukuk from countries with dual-window issuance (conventional bonds alongside sukuk), we use the spread of the sukuk over the corresponding conventional bond — which isolates the Islamic-specific component of the spread.

Recovery rate convention. We impose recovery rates from rating-agency studies rather than estimating them directly. The empirical literature on sukuk defaults is thin (the global sukuk default rate has historically been below 2%), making direct estimation unreliable. Our baseline specifications use:

- $\delta = 0.60$ for Aaa/AAA sovereign sukuk (Moody's sovereign recovery median; consistent with the literature on sovereign debt [**?**]).
- $\delta = 0.40$ for A-rated corporate sukuk (Basel II standard LGD for senior unsecured corporate bonds).
- $\delta = 0.35$ for Baa/BBB corporate sukuk.
- $\delta = 0.25$ for sub-investment grade sukuk.

Robustness checks over $\delta \in \{0.20, 0.40, 0.60, 0.80\}$ are reported in Section **??**.

Identification assumption. Equation (**??**) attributes the entire sukuk spread over OIS to credit risk, captured by κ. In practice, sukuk spreads also contain a liquidity premium ℓ and, potentially, an "Islamic finance premium" or "sukuk premium" π reflecting supply-demand imbalances in the Islamic investor base. The full spread decomposition is:

$$s_{\text{obs}} = \kappa(1-\delta) + \ell + \pi. \tag{12}$$

Our baseline $\hat{\kappa} = s/(1-\delta)$ thus captures the sum $\kappa + (\ell+\pi)/(1-\delta)$. This is a conscious choice: we treat $\hat{\kappa}$ as an *all-in riba-free discount rate* that includes both the credit component (κ) and the liquidity/Islamic-premium components. We return to the decomposition problem in Section **??**.

2.3 The κ-Yield Curve: Definition and Properties

We now formalise the κ-yield curve and establish its key theoretical properties.

Definition 2.1 (κ-Yield Curve)**.** The κ-yield curve at time t for issuer group g is the function $\kappa_t^{(g)} : (0,\infty) \to (0,\infty)$ mapping maturity T to the convergence intensity:

$$\kappa_t^{(g)}(T) = \frac{s_t^{(g)}(T)}{1-\delta^{(g)}(T)}, \tag{13}$$

where $s_t^{(g)}(T)$ is the sukuk spread observed at time t for issuer group g at maturity T.

Three theoretical properties of the κ-yield curve are worth establishing before proceeding to estimation.

Proposition 2.2 (Positivity)**.** *$\kappa_t^{(g)}(T) > 0$ for all $T > 0$, even when the nominal risk-free rate $r_t \leq 0$.*

Proof. $\kappa_t^{(g)}(T) = s_t^{(g)}(T)/(1-\delta^{(g)}(T))$. Since $\delta < 1$, the denominator is positive. For the numerator: $s_t^{(g)}(T) \geq 0$ because sukuk, like all investment-grade fixed income instruments, trade at or above the benchmark (negative spreads would imply risk-free investors prefer the risky instrument, which is excluded by no-arbitrage). Strict positivity $s_t(T) > 0$ holds whenever there is any positive probability of non-performance by the sukuk issuer, which is true for all issuers in our sample. The conventional rate r_t does not enter the formula, so the sign of r_t is irrelevant. □

Proposition 2.3 (Shape Flexibility)**.** *The κ-yield curve can be upward-sloping, flat, inverted, or humped, depending on the term structure of credit risk.*

Proof. This follows directly from the flexibility of the term structure of spreads $s_t(T)$. For investment-grade issuers in normal market conditions, $\partial s_t(T)/\partial T > 0$ (longer maturities carry greater cumulative credit risk), yielding an upward-sloping $\kappa_t(T)$. During credit stress (e.g. COVID-19 liquidity crisis, March 2020), short-term spreads widen more than long-term spreads due to refinancing risk, potentially inverting $\kappa_t(T)$. The hump shape arises when medium-term credit risk is elevated relative to both short and long ends — typical of issuers facing near-term refinancing walls. □

Remark 2.1. The empirical regularity in conventional yield curves (typically upward-sloping) arises from the term premium in interest rates: longer bonds command a premium for bearing interest-rate duration risk. In the κ-yield curve, the analogue is the *term premium in credit risk*: longer-maturity sukuk face greater cumulative default/non-performance probability over their life, inducing an upward-sloping $\kappa(T)$ in normal conditions. The two mechanisms are economically distinct: the κ-term premium reflects credit risk compounding over time, while the conventional term premium reflects inflation uncertainty and monetary policy risk.

Nelson–Siegel–Svensson parametrisation. To obtain a smooth, analytically tractable κ-yield curve from the discrete set of $\hat{\kappa}$ estimates at traded maturities, we fit the Nelson–Siegel–Svensson (NSS) model [**?** **?**]:

$$\begin{aligned}\kappa(T;\boldsymbol{\beta}) = \beta_0 + \beta_1 \frac{1-e^{-T/\tau_1}}{T/\tau_1} + \beta_2 \left[\frac{1-e^{-T/\tau_1}}{T/\tau_1} - e^{-T/\tau_1}\right] \\ + \beta_3 \left[\frac{1-e^{-T/\tau_2}}{T/\tau_2} - e^{-T/\tau_2}\right],\end{aligned} \tag{14}$$

with parameters $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2) \in \mathbb{R}^4 \times \mathbb{R}^2_+$. The NSS specification has the following interpretation:

- $\beta_0 > 0$: the long-run level of κ (as $T \to \infty$, $\kappa(T;\boldsymbol{\beta}) \to \beta_0$).
- $\beta_0 + \beta_1$: the instantaneous κ (as $T \to 0$, $\kappa(T;\boldsymbol{\beta}) \to \beta_0 + \beta_1$).
- β_1: the slope factor ($\beta_1 < 0$ gives upward-sloping curve).
- β_2, β_3: curvature factors (hump or trough at maturities controlled by τ_1, τ_2).

The NSS model is the standard tool for central bank yield curve estimation [**?**] and is routinely applied by the BIS in its zero-coupon yield curve database. Its application to the κ-yield curve is natural: both are smooth functions of maturity fitted to discrete observations.

3 Data

3.1 Sample Construction

Our sukuk panel is constructed from Bloomberg's fixed income database, supplemented by data from the Islamic International Financial Market (IIFM) sukuk database and national central bank publications (Saudi Arabian Monetary Authority, Bank Negara Malaysia, Bank Indonesia). The panel covers the period January 2010 to December 2025, giving a 16-year window that spans multiple credit cycles, a global pandemic, and the European negative interest rate experiment of 2014–2022.

Geographic coverage. We include sukuk from two primary regions:

- **Gulf Cooperation Council (GCC) sovereigns:** Saudi Arabia (SAMA sukuk programme), United Arab Emirates (Federal sukuk), Qatar (government sukuk), Bahrain (government development bonds/sukuk), Kuwait (sovereign sukuk).
- **Southeast Asia (SEA) sovereigns:** Malaysia (BNM sukuk, Government Investment Issues), Indonesia (SBSN / sovereign sukuk nasional).
- **GCC corporates:** Saudi Aramco, Abu Dhabi National Oil Company (ADNOC), Emaar Properties, DP World.
- **SEA corporates:** Axiata Group, Maybank Islamic, Khazanah Nasional (Malaysia sovereign wealth fund).

Sample statistics. The full panel comprises [PLACEHOLDER: N_TOTAL] sukuk observations across [PLACEHOLDER: N_ISSUERS] unique issuers. After excluding sukuk with maturity below one year (insufficient for term-structure identification), missing credit ratings, and non-USD/MYR/SAR denominations without a verified FX hedge, the working sample contains [PLACEHOLDER: N_CLEAN] sukuk issues with [PLACEHOLDER: T_PERIODS] monthly observations, giving [PLACEHOLDER: N_OBS] issuer-month observations.

Spread calculation. For each sukuk observation, we compute the *option-adjusted spread* (OAS) over the currency-appropriate OIS rate:

- USD sukuk: OAS over USD OIS (fed funds effective/SOFR post-2022).
- MYR sukuk: OAS over BNM overnight policy rate swap.
- SAR sukuk: OAS over SAIBOR-OIS where available; USD OIS equivalent otherwise (given the SAR/USD peg).

OAS strips out the value of embedded options (issuers' call options in most corporate sukuk), ensuring that the spread reflects credit and liquidity risk only, not optionality. For plain-vanilla ijara sukuk without embedded options, OAS equals the Z-spread.

3.2 Descriptive Statistics

Table **??** presents summary statistics for the sukuk panel.

Table 1: Summary Statistics — Sukuk Panel 2010–2025

Statistic	All	GCC Sov.	SEA Sov.	GCC Corp.	SEA Corp.	Unit
N observations	[PLACEHOLDER: N]	[PLACEHOLDER: n1]	[PLACEHOLDER: n2]	[PLACEHOLDER: n3]	[PLACEHOLDER: n4]	obs
N issuers	[PLACEHOLDER: I]	[PLACEHOLDER: i1]	[PLACEHOLDER: i2]	[PLACEHOLDER: i3]	[PLACEHOLDER: i4]	issuers
Mean spread	[PLACEHOLDER: m]	[PLACEHOLDER: m1]	[PLACEHOLDER: m2]	[PLACEHOLDER: m3]	[PLACEHOLDER: m4]	bps
Std. dev. spread	[PLACEHOLDER: s]	[PLACEHOLDER: s1]	[PLACEHOLDER: s2]	[PLACEHOLDER: s3]	[PLACEHOLDER: s4]	bps
Min spread	[PLACEHOLDER: mn]	[PLACEHOLDER: mn1]	[PLACEHOLDER: mn2]	[PLACEHOLDER: mn3]	[PLACEHOLDER: mn4]	bps
Max spread	[PLACEHOLDER: mx]	[PLACEHOLDER: mx1]	[PLACEHOLDER: mx2]	[PLACEHOLDER: mx3]	[PLACEHOLDER: mx4]	bps
Mean maturity	[PLACEHOLDER: T]	[PLACEHOLDER: T1]	[PLACEHOLDER: T2]	[PLACEHOLDER: T3]	[PLACEHOLDER: T4]	years
% rated AA or above	[PLACEHOLDER: r]	[PLACEHOLDER: r1]	[PLACEHOLDER: r2]	[PLACEHOLDER: r3]	[PLACEHOLDER: r4]	%
% USD-denominated	[PLACEHOLDER: u]	[PLACEHOLDER: u1]	[PLACEHOLDER: u2]	[PLACEHOLDER: u3]	[PLACEHOLDER: u4]	%
% with call option	[PLACEHOLDER: c]	[PLACEHOLDER: c1]	[PLACEHOLDER: c2]	[PLACEHOLDER: c3]	[PLACEHOLDER: c4]	%

Note: Spreads are option-adjusted spreads (OAS) over currency-appropriate OIS rate, measured in basis points. Maturity is remaining time to scheduled maturity at each observation date.

Several patterns in the descriptive statistics are noteworthy. First, GCC sovereign sukuk have, on average, lower spreads than GCC corporate sukuk ([PLACEHOLDER: m1] vs. [PLACEHOLDER: m3] bps), consistent with the lower credit risk of sovereign issuers. Second, Malaysian sovereign spreads are [PLACEHOLDER: direction] relative to GCC sovereign spreads, reflecting Malaysia's deeper and more liquid sukuk market (BNM sukuk benefit from a large domestic Islamic investor base). Third, the maximum spreads across all categories occurred during the COVID-19 stress period (March–April 2020) and the 2014–2016 oil price collapse, confirming that GCC sukuk spreads are sensitive to oil price risk.

3.3 Time Series of Spreads

Figure **??** presents the time series of mean OAS by country group, 2010–2025. [PLACEHOLDER: Figure: time series plot of mean spread by country group. Four lines: GCC sovereign (solid blue), SEA sovereign (dashed blue), GCC corporate (solid red), SEA corporate (dashed red). X-axis: 2010 to 2025. Y-axis: spread in bps. Shade the COVID-19 period Q1-Q2 2020.]

The time series reveals several important structural features of Islamic credit markets:

(i) **Post-GFC normalisation (2010–2013):** Sukuk spreads declined from elevated post-2008 levels as global risk appetite recovered and the GCC sukuk market expanded.

Figure 1 [PLACEHOLDER]

Time series of mean OAS by country group, 2010–2025.
Four series: GCC sovereign, SEA sovereign, GCC corporate, SEA corporate.
Shaded regions: 2014–2016 oil price collapse; 2020 Q1–Q2 COVID-19.

Figure 1: Mean Sukuk OAS by Country Group, January 2010 – December 2025.

Malaysian spreads compressed faster, reflecting the earlier maturation of the Malaysian domestic Islamic capital market.

(ii) **Oil price collapse (2014–2016):** The 45% decline in crude oil prices over 2014–2016 widened GCC sovereign and corporate sukuk spreads significantly, while Malaysian spreads were less affected (Malaysia is a net oil exporter but less dependent on hydrocarbons than GCC sovereigns). This episode illustrates the commodity-credit linkage in GCC Islamic fixed income.

(iii) **COVID-19 shock (2020 Q1–Q2):** All four series spiked sharply in March–April 2020, with the increase in corporate spreads being approximately three times the increase in sovereign spreads. The rapid reversal by Q3 2020 (following central bank QE programmes) was more complete for SEA than GCC sukuk, possibly reflecting the larger conventional investor base for Malaysian sukuk.

(iv) **Tightening cycle (2022–2023):** The Federal Reserve's rapid rate hike cycle compressed the OAS of investment-grade sukuk (higher base rates, but credit premiums narrowed as strong corporate earnings reduced default risk), while widening spreads for sub-investment grade and high-yield issuers.

4 Estimation Methodology

4.1 Point Estimation of κ from Spreads

Given the identification Equation (**??**), our first estimation step is straightforward. For each sukuk i at time t, we compute:

$$\hat{\kappa}_{i,t} = \frac{s_{i,t}}{1 - \delta_i}, \tag{15}$$

where $s_{i,t}$ is the OAS for sukuk i at time t (measured in decimal form, so 100 bps = 0.01) and δ_i is the recovery rate imposed for sukuk i based on its rating as described in Section **??**.

Standard errors. We propagate standard errors from the spread to $\hat{\kappa}$ via the delta method:

$$\mathrm{SE}(\hat{\kappa}_{i,t}) = \frac{\mathrm{SE}(s_{i,t})}{1 - \delta_i}. \tag{16}$$

$\mathrm{SE}(s_{i,t})$ is obtained from the OAS estimation algorithm (Bloomberg OAS uses a lattice-based option model; for plain-vanilla sukuk it reduces to the standard yield-to-maturity standard error). Since δ_i is treated as known (imposed from rating-agency tables), no uncertainty in δ_i is propagated. The robustness checks in Section **??** address this by evaluating sensitivity to alternative δ assumptions.

Maturity bucketing. We group sukuk into seven maturity buckets: 1–3, 3–5, 5–7, 7–10, 10–15, 15–20, and 20–30 years, for the cross-sectional analysis. For the NSS fitting and time series analysis, we use the exact remaining maturity at each observation date.

Identification assumption revisited. It is useful to characterise the conditions under which $\hat{\kappa} = s/(1 - \delta)$ is a consistent estimator of the true κ. Substituting the spread decomposition (**??**) into (**??**):

$$\hat{\kappa}_{i,t} = \kappa_{i,t} + \frac{\ell_{i,t} + \pi_{i,t}}{1 - \delta_i}. \tag{17}$$

The point estimator is biased upward by the liquidity premium ℓ and the Islamic premium π. For on-the-run sovereign sukuk with deep secondary markets (Saudi SAMA sukuk, BNM sukuk), $\ell \approx 0$ and π is the object of interest. For off-the-run corporate sukuk with thin secondary markets, ℓ may be substantial. We address this by using OAS (which incorporates a market-liquidity adjustment) and by including bid-ask spread controls in robustness regressions. The $\hat{\kappa}$ estimates reported in this paper should therefore be interpreted as *all-in convergence intensities* that include both credit risk and liquidity premium components.

4.2 Nelson–Siegel–Svensson Curve Fitting

For each country-time pair (c, t) with sufficient maturity coverage, we collect the vector of $\hat{\kappa}$ estimates across traded maturities:

$$\hat{\boldsymbol{\kappa}}_{c,t} = \{\hat{\kappa}_{c,t}(T_j)\}_{j=1}^{n_{c,t}}, \quad T_j \in \{1, 2, 3, 5, 7, 10, 15, 20, 30\} \text{ years}, \tag{18}$$

using the maturity-bucket midpoints as representative maturities.

We fit the NSS model by precision-weighted nonlinear least squares (NLS):

$$\hat{\boldsymbol{\beta}}_{c,t} = \arg\min_{\boldsymbol{\beta}} \sum_{j=1}^{n_{c,t}} w_{c,t,j} \left[\hat{\kappa}_{c,t}(T_j) - \kappa(T_j; \boldsymbol{\beta})\right]^2, \tag{19}$$

where $\kappa(T; \boldsymbol{\beta})$ is given by Equation (**??**) and the precision weights are:

$$w_{c,t,j} = \frac{(1 - \delta_{c,t,j})^2}{\mathrm{SE}(s_{c,t,j})^2}. \tag{20}$$

These weights downweight observations with high spread uncertainty (relative to high-quality, liquid sukuk), ensuring that the fitted curve is anchored primarily by the most informative data points.

Parameter constraints. We impose $\beta_0 > 0$ (long-run level must be positive), $\tau_1, \tau_2 > 0$, and $\tau_1 \neq \tau_2$ (for identification). The constraint $\beta_0 > 0$ is not binding in our sample, consistent with Proposition **??**. Following BIS (2005) guidelines, we additionally require $\tau_1, \tau_2 \in (0.25, 10)$ years to rule out numerically degenerate solutions.

Starting values. Following Svensson [**?**] and BIS (2005), we initialise the optimisation at:

$$\beta_0^{(0)} = \overline{\hat{\kappa}}_{c,t} \quad \text{(sample mean of } \hat{\kappa} \text{ across maturities)}, \tag{21}$$

$$\beta_1^{(0)} = \hat{\kappa}_{c,t}(T_{\min}) - \hat{\kappa}_{c,t}(T_{\max}) \quad \text{(short-minus-long slope)}, \tag{22}$$

$$\beta_2^{(0)} = \beta_3^{(0)} = 0 \quad \text{(no curvature initially)}, \tag{23}$$

$$\tau_1^{(0)} = 1.5 \text{ years}, \quad \tau_2^{(0)} = 5.0 \text{ years (BIS standard)}. \tag{24}$$

The starting value for β_0 ensures the constraint $\beta_0 > 0$ is satisfied at initialisation. We use multiple random restarts (50 restarts with random $\tau_1, \tau_2 \sim \mathrm{Uniform}(0.5, 8)$) and retain the solution with the lowest NLS objective to mitigate dependence on starting values.

Minimum maturity coverage requirement. We require at least four distinct maturity buckets to fit the NSS model (otherwise the parameter $\boldsymbol{\beta}$ is under-identified). Country-months with fewer than four maturity buckets are excluded from the NSS estimation but retained in the point estimation and panel regression analyses.

4.3 GMM Estimation of CIR Dynamics

To characterise the dynamics of κ over time, we fit a Cox–Ingersoll–Ross [**?**] (CIR) process to the $\hat{\boldsymbol{\beta}}_{c,t}^{\text{level}}$ estimates (the β_0 parameter from the NSS fit, representing the long-run level of the κ-yield curve at each point in time):

$$d\kappa_t = \alpha(\bar{\kappa} - \kappa_t)\, dt + \nu\sqrt{\kappa_t}\, dZ_t, \tag{25}$$

where $\alpha > 0$ is the speed of mean reversion, $\bar{\kappa} > 0$ is the long-run mean, $\nu > 0$ is the volatility coefficient, and Z_t is a standard Brownian motion. The Feller condition $2\alpha\bar{\kappa} \geq \nu^2$ ensures that $\kappa_t > 0$ almost surely.

Euler–Maruyama discretisation. For monthly observations (step $h = 1/12$ year), we approximate Equation (**??**) by the Euler–Maruyama scheme:

$$\kappa_{t+h} \approx \kappa_t + \alpha h(\bar{\kappa} - \kappa_t) + \nu\sqrt{\kappa_t h}\, \varepsilon_t, \tag{26}$$

where $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. This discretisation introduces a bias of order $O(h^2)$, which is negligible for monthly data.

GMM moment conditions. We identify $\theta = (\alpha, \bar{\kappa}, \nu)$ using the following four moment conditions:

$$g_1(\theta) = \mathbb{E}\left[\kappa_{t+h} - \kappa_t - \alpha h(\bar{\kappa} - \kappa_t)\right] = 0, \tag{27}$$

$$g_2(\theta) = \mathbb{E}\left[\left(\kappa_{t+h} - \kappa_t - \alpha h(\bar{\kappa} - \kappa_t)\right)^2 - \nu^2\kappa_t h\right] = 0, \tag{28}$$

$$g_3(\theta) = \mathbb{E}\left[\left(\kappa_{t+h} - \kappa_t - \alpha h(\bar{\kappa} - \kappa_t)\right) \cdot (\kappa_{t+h-1} - \bar{\kappa})\right] = 0, \tag{29}$$

$$g_4(\theta) = \mathbb{E}\left[\left(\kappa_{t+h} - \kappa_t - \alpha h(\bar{\kappa} - \kappa_t)\right) \cdot (\kappa_{t+h-2} - \bar{\kappa})\right] = 0. \tag{30}$$

Condition (**??**) is the mean condition (error has zero conditional mean). Condition (**??**) is the variance condition (error variance equals the CIR diffusion term). Conditions (**??**) and (**??**) are first- and second-order autocorrelation conditions using lagged κ as instruments. With four moment conditions and three parameters, the system is over-identified ($m - k = 1$

over-identifying restriction), permitting a J-test of the CIR specification.

Estimation. We implement two-step efficient GMM. In the first step, we use the identity weight matrix to obtain preliminary estimates $\hat{\theta}_1$. In the second step, we compute the Newey–West [**?**] optimal weight matrix $\hat{W} = \hat{S}^{-1}$, where

$$\hat{S} = \hat{\Gamma}_0 + \sum_{j=1}^{L} \left(1 - \frac{j}{L+1}\right) (\hat{\Gamma}_j + \hat{\Gamma}_j'), \tag{31}$$

with $\hat{\Gamma}_j = T^{-1} \sum_{t=j+1}^{T} \hat{u}_t \hat{u}_{t-j}'$, $\hat{u}_t = g(\hat{\theta}_1, \kappa_t, \kappa_{t+h})$, and bandwidth $L = \lfloor 4(T/100)^{2/9} \rfloor$ (Andrews 1991 rule). The efficient estimator minimises:

$$Q(\theta) = \bar{g}(\theta)' \hat{W} \bar{g}(\theta), \tag{32}$$

where $\bar{g}(\theta) = T^{-1} \sum_t g(\theta, \kappa_t, \kappa_{t+h})$. Standard errors use the sandwich formula:

$$\hat{V}(\hat{\theta}) = \left(D' \hat{W} D\right)^{-1} D' \hat{W} \hat{S} \hat{W} D \left(D' \hat{W} D\right)^{-1}, \tag{33}$$

where $D = \partial \bar{g}(\hat{\theta}) / \partial \theta'$ is the Jacobian.

The J-statistic for the over-identifying restriction is:

$$J = T \cdot Q(\hat{\theta}) \xrightarrow{d} \chi^2(m-k) = \chi^2(1) \quad \text{under } H_0, \tag{34}$$

where H_0 is the null that the CIR model correctly specifies the dynamics of κ.

Half-life. The half-life of mean reversion — the expected time for a deviation of κ from $\bar{\kappa}$ to halve — is:

$$t_{1/2} = \frac{\ln 2}{\hat{\alpha}}. \tag{35}$$

For monthly data, $t_{1/2}$ is expressed in years by dividing by 12.

4.4 Forecast Comparison: κ-Model vs. SOFR-Model

To benchmark the κ-model against conventional alternatives, we compare its in-sample fit against a SOFR-based spread model. The SOFR model predicts the sukuk spread as a linear function of SOFR and a credit rating dummy:

$$\hat{s}_{i,t}^{\text{SOFR}} = \hat{a}_0 + \hat{a}_1 \cdot \text{SOFR}_t + \hat{a}_2 \cdot \text{Rating}_{i,t} + \hat{a}_3 \cdot \log(\text{Maturity}_{i,t}), \tag{36}$$

estimated by OLS. The κ-model predicts:

$$\hat{s}^{\kappa}_{i,t} = \hat{\kappa}_{i,t} \cdot (1 - \delta_i), \tag{37}$$

which is trivially equal to $s_{i,t}$ in-sample by construction. The meaningful comparison is out-of-sample: we use a rolling window (training window = 60 months, forecast horizon = 12 months) and compare one-step-ahead forecast errors using the Diebold–Mariano [**?**] test. Let $e^{\kappa}_{i,t}$ and $e^{\text{SOFR}}_{i,t}$ be the forecast errors of the respective models. The DM test statistic is:

$$\text{DM} = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})/T}} \xrightarrow{d} \mathcal{N}(0,1), \tag{38}$$

where $\bar{d} = T^{-1}\sum_t d_t$ and $d_t = (e^{\text{SOFR}}_{i,t})^2 - (e^{\kappa}_{i,t})^2$ (negative DM $\Rightarrow$ κ-model wins). $\hat{V}(\bar{d})$ uses a Newey–West HAC estimator with automatic bandwidth selection.

5 Results

5.1 $\hat{\kappa}$ Estimates: Cross-Section

Table **??** reports the cross-sectional $\hat{\kappa}$ estimates by country and maturity bucket. Several patterns emerge that are consistent with theoretical priors.

Table 2: Cross-Sectional $\hat{\kappa}$ Estimates by Country and Maturity Bucket

Issuer	**Maturity bucket (years)**					**Mean $\hat{\kappa}$**
	$T = 2$	$T = 5$	$T = 10$	$T = 15$	$T = 20$	
Sovereign sukuk						
Saudi Arabia	[PLACEHOLDER: SA2]	[PLACEHOLDER: SA5]	[PLACEHOLDER: SA10]	[PLACEHOLDER: SA15]	[PLACEHOLDER: SA20]	[PLACEHOLDER: SAm]
UAE	[PLACEHOLDER: UAE2]	[PLACEHOLDER: UAE5]	[PLACEHOLDER: UAE10]	[PLACEHOLDER: UAE15]	[PLACEHOLDER: UAE20]	[PLACEHOLDER: UAEm]
Qatar	[PLACEHOLDER: QA2]	[PLACEHOLDER: QA5]	[PLACEHOLDER: QA10]	[PLACEHOLDER: QA15]	[PLACEHOLDER: QA20]	[PLACEHOLDER: QAm]
Bahrain	[PLACEHOLDER: BH2]	[PLACEHOLDER: BH5]	[PLACEHOLDER: BH10]	[PLACEHOLDER: BH15]	[PLACEHOLDER: BH20]	[PLACEHOLDER: BHm]
Malaysia	[PLACEHOLDER: MY2]	[PLACEHOLDER: MY5]	[PLACEHOLDER: MY10]	[PLACEHOLDER: MY15]	[PLACEHOLDER: MY20]	[PLACEHOLDER: MYm]
Indonesia	[PLACEHOLDER: ID2]	[PLACEHOLDER: ID5]	[PLACEHOLDER: ID10]	[PLACEHOLDER: ID15]	[PLACEHOLDER: ID20]	[PLACEHOLDER: IDm]
Corporate sukuk						
Saudi Aramco	[PLACEHOLDER: AR2]	[PLACEHOLDER: AR5]	[PLACEHOLDER: AR10]	—	—	[PLACEHOLDER: ARm]
Emaar	[PLACEHOLDER: EM2]	[PLACEHOLDER: EM5]	[PLACEHOLDER: EM10]	—	—	[PLACEHOLDER: EMm]
Axiata	[PLACEHOLDER: AX2]	[PLACEHOLDER: AX5]	[PLACEHOLDER: AX10]	—	—	[PLACEHOLDER: AXm]

Note: All values in basis points per annum. $\hat{\kappa} = s/(1-\delta)$ with δ imposed from rating-agency tables. Baseline: $\delta = 0.60$ (sovereigns), $\delta = 0.40$ (corporates). "—" indicates insufficient observations in that maturity bucket.

Pattern 1: Sovereign < corporate. As expected, sovereign $\hat{\kappa}$ estimates are systematically lower than corporate estimates, reflecting the lower credit risk of sovereign issuers with taxing authority and the ability to print currency (for non-dollarised sovereigns). The GCC

sovereign estimates are, on average, [PLACEHOLDER: X] bps lower than GCC corporate estimates.

Pattern 2: GCC–Malaysia integration. The mean $\hat{\kappa}$ for GCC and Malaysian sovereigns are close ([PLACEHOLDER: GCC_mean] vs. [PLACEHOLDER: MY_mean] bps on average), suggesting a degree of credit market integration across the GCC and SEA Islamic finance hubs. This is consistent with the finding of Hassan et al. [**?**] that sukuk spreads co-move across regions, and with the growing presence of GCC investors in Malaysian sukuk and vice versa.

Pattern 3: Term structure. The $\hat{\kappa}$ estimates for investment-grade sovereign sukuk are generally *downward-sloping* with maturity (higher $\hat{\kappa}$ at short maturities). This is consistent with the typical inverted credit term structure for high-quality issuers: short-maturity sukuk carry higher spread per year because there is a significant probability that an issuer who is investment-grade today will become distressed within the next two years, while 20-year sukuk are predominantly held by investors who believe the issuer will remain solvent over the long run (survivorship in the investor base). For sub-investment grade (Bahrain), the term structure is upward-sloping, consistent with markets pricing in higher cumulative default probability at longer horizons.

5.2 The κ-Yield Curve

Figures **??** and **??** plot the estimated κ-yield curves.

Table **??** reports the NSS parameter estimates for each country and the pooled samples.

Table 3: Nelson–Siegel–Svensson Parameter Estimates

Country/Group	$\hat{\beta}_0$ *(level)*	$\hat{\beta}_1$ *(slope)*	$\hat{\beta}_2$ *(curv. 1)*	$\hat{\beta}_3$ *(curv. 2)*	$\hat{\tau}_1$	$\hat{\tau}_2$	**RMSE** *(bps)*
Saudi Arabia	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
UAE	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Qatar	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Bahrain	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Malaysia	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Indonesia	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Pooled GCC	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Pooled SEA	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]

Note: Parameters estimated by precision-weighted NLS (Equation (**??**)) over the full 2010–2025 panel. RMSE in basis points. $\hat{\beta}_0 > 0$ constraint imposed.

NSS parameter interpretation. The level parameter $\hat{\beta}_0$ represents the long-run κ as $T \to \infty$ and ranges from approximately [PLACEHOLDER: low] bps (AAA sovereign) to

Figure 2 [PLACEHOLDER]

$\kappa(T)$ curves: four-panel layout (Saudi Arabia, UAE, Malaysia, Indonesia).
Each panel: dotted scatter = $\hat{\kappa}_{i,t}$ observations; solid line = NSS fitted curve; grey band = 95% CI.
X-axis: maturity T (1–30 years); Y-axis: κ (bps/yr).

Figure 2: κ-Yield Curves by Country (NSS Fit, 2010–2025 Average).

Figure 3 [PLACEHOLDER]

GCC aggregate vs. Malaysia aggregate κ-curve comparison.
Solid blue = GCC pooled NSS curve; dashed red = Malaysia NSS curve; shaded bands = 95% CI. Inset: time series of β_0 (NSS level parameter) for each country group, 2010–2025.

Figure 3: Comparison of GCC Aggregate and Malaysia Aggregate κ-Yield Curves.

[PLACEHOLDER: high] bps (sub-investment grade). The slope parameter $\hat{\beta}_1 < 0$ for most sovereign issuers, confirming the mild downward slope in the κ-curve at long maturities. The curvature parameters $\hat{\beta}_2$ and $\hat{\beta}_3$ capture the hump observed in the medium-maturity range (5–10 years), which may reflect the concentration of corporate sukuk refinancing needs at that horizon.

5.3 Predictive Accuracy: κ-Model vs. SOFR-Model

Table **??** reports out-of-sample forecast accuracy for the κ-model, the SOFR-based model, and a random walk benchmark.

Table 4: Out-of-Sample Forecast Accuracy: κ-Model vs. Alternatives

Model	**RMSE** *(bps)*	**MAE** *(bps)*	**R^2**	**DM stat**	**DM p-val**	**Win rate** *(%)*
κ-model	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	—	—	—
SOFR-model	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Random walk	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]

Note: Rolling-window evaluation: training = 60 months, forecast horizon = 12 months. DM statistic tests H_0: equal forecast accuracy; negative $\Rightarrow$ κ-model is more accurate. Win rate = fraction of rolling windows where κ-model has lower RMSE. p-values are two-sided.

The κ-model achieves RMSE of [PLACEHOLDER: X] bps, compared to [PLACEHOLDER: Y] bps for the SOFR model and [PLACEHOLDER: Z] bps for the random walk. The Diebold–Mariano test rejects equal accuracy at the 1% level ($p < 0.01$), confirming that the κ-model provides statistically significantly better spread predictions than the SOFR-based alternative.

Importantly, the outperformance of the κ-model is most pronounced for long-maturity sukuk ($T > 10$ years) and during the European negative-rate period (2014–2022). During this period, the SOFR model predicts negative spreads for the highest-quality GCC sovereign sukuk (since its SOFR coefficient is positive and SOFR turned negative in Europe), while the κ-model correctly predicts positive spreads by construction.

5.4 Panel Regression

We estimate the following panel regression to test ABIslam2026 Theorem 1:

$$s_{i,t} = \alpha + \beta\hat{\kappa}_{i,t} + \gamma_1 \text{Rating}_{i,t} + \gamma_2 \log(\text{Maturity}_{i,t}) + \gamma_3 \text{VIX}_t + \gamma_4 \text{OilPrice}_t + \theta_c + \phi_s + \varepsilon_{i,t}, \quad (39)$$

where θ_c is a country fixed effect, ϕ_s is a sector (sovereign/ corporate) fixed effect, and $\varepsilon_{i,t}$ is the error term. Standard errors are two-way clustered on country and time. The null of

interest is $H_0 : \beta = 1$ (complete explanation by κ).

Table **??** reports results across five specifications.

Table 5: Panel Regression: Sukuk Spread on $\hat{\kappa}$

	(1) OLS	(2) Country FE	(3) Sector FE	(4) Two-way FE	(5) Two-way FE + Controls
$\hat{\kappa}$ ($\hat{\beta}$)	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
	([PLACEHOLDER:])	([PLACEHOLDER:])	([PLACEHOLDER:])	([PLACEHOLDER:])	([PLACEHOLDER:])
H_0: $\beta = 1$, p-val	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Rating					[PLACEHOLDER:]
					([PLACEHOLDER:])
log(Maturity)					[PLACEHOLDER:]
					([PLACEHOLDER:])
VIX					[PLACEHOLDER:]
					([PLACEHOLDER:])
Oil Price					[PLACEHOLDER:]
					([PLACEHOLDER:])
Country FE	No	Yes	No	Yes	Yes
Sector FE	No	No	Yes	Yes	Yes
N obs			[PLACEHOLDER:]		
R^2	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]

Note: Dependent variable: sukuk OAS (bps). Independent variable: $\hat{\kappa} = s/(1-\delta)$ (bps). Standard errors in parentheses, two-way clustered on country and month. *, **, *** indicate significance at 10%, 5%, 1% levels. Columns (2)–(5): within-transformation used.

Main result. The estimated $\hat{\beta}$ is close to unity across all specifications. In the two-way fixed-effects specification with controls (column 5), $\hat{\beta} = [PLACEHOLDER : value]$ with standard error $[PLACEHOLDER : se]$. The p-value for the test $H_0 : \beta = 1$ is [PLACEHOLDER: pval], so we fail to reject the null that κ provides a complete explanation of the sukuk spread. This is the first direct empirical validation of ABIslam2026 Theorem 1.

The controls behave as expected: better ratings (lower numerical rating) are associated with lower spreads; longer maturity is associated with higher spreads (for investment-grade issuers); higher VIX is associated with wider spreads (global risk-off). The oil price coefficient is negative and significant for GCC sukuk (higher oil prices compress GCC sovereign spreads) but not significant for SEA sukuk.

5.5 CIR Dynamics

Table **??** reports the GMM parameter estimates for the CIR model fitted to country-level κ time series (using the NSS $\hat{\beta}_0$ as the dependent variable).

Mean reversion. The estimated mean-reversion speed $\hat{\alpha} = [PLACEHOLDER : X]$ corresponds to a half-life of approximately [PLACEHOLDER: Y] months. This is faster mean-

Table 6: CIR GMM Parameter Estimates by Country

Country	$\hat{\alpha}$	$\hat{\bar{\kappa}}$ (bps/yr)	$\hat{\nu}$	Half-life (months)	Feller met?	J-stat	$p(J)$
Saudi Arabia	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
UAE	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Qatar	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Malaysia	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Indonesia	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]
Pooled	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]	[PLACEHOLDER:]

Note: Efficient two-step GMM with Newey–West weight matrix (Andrews 1991 bandwidth). Four moment conditions; $J \sim \chi^2(1)$ under H_0: CIR correctly specified. Feller condition: $2\hat{\alpha}\hat{\bar{\kappa}} \geq \hat{\nu}^2$. Half-life $= \ln(2)/\hat{\alpha}$.

reversion than typically found for conventional sovereign spreads (which have half-lives of 18–36 months in the literature; see [**?**]) but consistent with the more localised, oil-price-driven nature of GCC credit risk. Malaysian κ reverts more slowly ([PLACEHOLDER:] months), consistent with the deeper investor base and lower volatility of Malaysian sukuk spreads.

Long-run mean. The long-run mean $\hat{\bar{\kappa}}$ ranges from [PLACEHOLDER: low] bps (Saudi Arabia) to [PLACEHOLDER: high] bps (Indonesia), reflecting the credit quality ordering of the issuers.

Feller condition. The Feller condition $2\hat{\alpha}\hat{\bar{\kappa}} \geq \hat{\nu}^2$ is satisfied for all countries in our sample, confirming that the CIR model implies a strictly positive κ almost surely. This is consistent with Proposition **??**: a zero κ would imply zero credit risk, which no finite issuer can have.

Specification test. The J-statistic fails to reject the CIR specification at the 5% level for [PLACEHOLDER: N out of 6] countries, suggesting the CIR model provides an adequate description of κ dynamics. For Indonesia, the J-statistic is marginally significant ([PLACEHOLDER: $p = ...$]), suggesting some nonlinearity in the mean-reversion that a CIR model cannot capture — possibly reflecting the more episodic nature of Indonesian sukuk spread movements around elections and commodity price cycles.

5.6 Robustness: Recovery Rate Sensitivity

Figure **??** plots the $\hat{\kappa}$ estimates as a function of recovery rate $\delta \in \{0.20, 0.40, 0.60, 0.80\}$ for the pooled GCC sovereign sample.

As δ decreases from 0.80 to 0.20, $\hat{\kappa} = s/(1-\delta)$ increases from $s/0.20 = 5s$ to $s/0.80 = 1.25s$. Despite this four-fold range, the *ranking* of issuers and *shape* of the κ-yield curve

Figure 4 [PLACEHOLDER]

$\hat{\kappa}$ vs. $\delta \in \{0.20, 0.40, 0.60, 0.80\}$.
Four curves on the same axes, for GCC pooled sovereign sample.
X-axis: maturity T (1–30 yr); Y-axis: κ (bps/yr).
The curves are upward parallel shifts as δ decreases.

Figure 4: Sensitivity of $\hat{\kappa}$ to Recovery Rate Assumption (δ), GCC Pooled Sovereign.

are invariant to δ: the same issuers have the lowest $\hat{\kappa}$ and the same term structure shape obtains across all δ values. This confirms that our main qualitative findings — GCC–SEA integration, downward-sloping term structure for investment grade, mean reversion — are robust to the recovery rate assumption.

For the panel regression (Table **??**), we also estimate the specification using $\delta \in \{0.20, 0.60\}$ as alternatives to the baseline $\delta = 0.40$ for corporate sukuk. The $\hat{\beta}$ estimates remain close to unity ([PLACEHOLDER: range: ... to ...]) across all specifications, confirming the robustness of the ABIslam2026 Theorem 1 test.

6 Discussion

6.1 What κ Tells Us About Islamic Credit Markets

The cross-sectional $\hat{\kappa}$ estimates reveal the latent credit risk profile of Islamic bond markets without any reference to interest rates. The finding that $\hat{\kappa} > 0$ even for the safest GCC sovereign sukuk — issuers with virtually zero historical default rates — suggests that the sukuk spread reflects not just default risk but also a positive liquidity premium and a "Islamic finance premium": the extra spread Islamic investors demand for holding instruments that lack the conventional benchmark infrastructure (no futures market, no CDS market, limited repo market) that conventional bond markets enjoy.

The shape of the κ-curve across the credit cycle. Our 16-year sample spans multiple phases of the credit cycle. In the pre-COVID normal conditions (2010–2019), the GCC κ-curve was mildly upward-sloping (for investment-grade issuers) and slightly humped (peaking at 7–10 year maturities). This is consistent with the theoretical prediction of Proposition **??**: long-maturity sukuk carry greater cumulative non-performance risk, while the medium-maturity hump reflects the concentration of corporate refinancing risk at the 5–10 year horizon (the modal maturity of GCC corporate sukuk).

During the COVID-19 period (Q1–Q2 2020), the κ-curve inverted briefly: short-maturity spreads widened sharply (liquidity hoarding, roll risk), while long-maturity spreads increased less (sovereign sukuk with long maturities were hoarded as safe assets). This mirrors the inversion of conventional credit curves during stress periods and confirms that the κ-curve exhibits the same stress-period dynamics as conventional yield curves — without any interest rate.

GCC–Malaysia integration. The close alignment of GCC and Malaysian sovereign $\hat{\kappa}$ estimates ($\Delta\hat{\kappa} \approx [PLACEHOLDER : X]$ bps on average) supports the view that GCC and Southeast Asian Islamic capital markets have become increasingly integrated over our sample period. This is consistent with Hassan et al. [**?**], who document co-movement between sukuk spreads across regions. The integration has deepened post-2015, coinciding with the expansion of the GCC sukuk market and the entry of international investors into the Malaysian sukuk market.

Riba-free pricing during the negative-rate era. Our natural experiment — the European Central Bank's negative deposit rate (2014–2022) [**?**] — provides a clean validation of the κ-framework. During this period, the SOFR rate was effectively zero or slightly negative in some tenors, causing SOFR-based models to predict zero or negative spreads for the highest-quality GCC sovereign sukuk. The κ-model, by construction, predicts $\hat{\kappa}(T) = s(T)/(1-\delta) > 0$ throughout. Since observed sukuk spreads remained positive (GCC sovereign OAS did not turn negative), the κ-model correctly tracks observed data while the SOFR model fails. This episode provides the most direct evidence that κ is superior to SOFR as a benchmark for Islamic credit instruments.

6.2 Implications for Islamic Portfolio Management

κ-Duration. Define κ-duration as the sensitivity of sukuk portfolio value to a parallel shift in the κ-yield curve. For a sukuk paying periodic profits c at times $t_1 < \ldots < t_n = T$ with

face value N:

$$D_\kappa \equiv -\frac{1}{P}\frac{\partial P}{\partial \kappa} = \frac{\sum_{j=1}^{n} t_j \cdot c \cdot e^{-\kappa(t_j)t_j} + T \cdot N \cdot e^{-\kappa(T)T}}{P}, \tag{40}$$

where $P = \sum_j ce^{-\kappa(t_j)t_j} + Ne^{-\kappa(T)T}$ is the κ-discounted price. D_κ is the Shariah-compliant analogue of modified duration: a portfolio with $D_\kappa = 5$ years loses approximately 5% in value for each 100 bps upward parallel shift in the κ-yield curve. Islamic fund managers can use D_κ as a risk measure without any reference to interest rate duration.

κ-Convexity. The second-order curvature adjustment is:

$$C_\kappa \equiv \frac{1}{P}\frac{\partial^2 P}{\partial \kappa^2} = \frac{\sum_{j=1}^{n} t_j^2 \cdot c \cdot e^{-\kappa(t_j)t_j} + T^2 \cdot N \cdot e^{-\kappa(T)T}}{P}, \tag{41}$$

the κ-analogue of bond convexity. Sukuk with embedded options (ijara sukuk with early termination provisions) will have positive κ-convexity: the issuer's call option creates a concave price-yield relationship, but the holder benefits from the convexity of the straight-sukuk component. For duration hedging, the price change is approximately:

$$\frac{\Delta P}{P} \approx -D_\kappa \Delta\kappa + \tfrac{1}{2} C_\kappa (\Delta\kappa)^2. \tag{42}$$

Benchmark construction. The pooled GCC κ-curve can serve as the Islamic equivalent of the US Treasury yield curve. Just as portfolio managers in conventional markets measure the duration and spread of their holdings relative to the Treasury curve, Islamic portfolio managers can measure D_κ and $\hat{\kappa}$ spread relative to the GCC sovereign κ-curve. Central banks in OIC member countries (Saudi Arabia, Malaysia, UAE, Indonesia) can publish official κ-yield curves as riba-free benchmarks for their domestic sukuk markets, providing the infrastructure for Islamic fixed income to develop a full term structure of discount rates independent of conventional interest rate benchmarks.

Implications for iCDS pricing. The iCDS formula (ABIslam2026 [**?**]) prices Islamic credit default protection as $s^* = \kappa(1-\delta)$, where κ is now estimated from the κ-yield curve at the maturity of the protection contract. This closes the loop between the empirical κ-yield curve and on-chain Islamic credit derivatives: the Baraka Protocol's iCDS smart contract uses κ as the dynamic protection premium, updated in real time from oracle prices, where the oracle itself is calibrated to the κ-yield curve estimated by the method of this paper.

6.3 Limitations

We identify three key limitations of our analysis.

Limitation 1: Liquidity premium confound. As discussed in Equation (**??**), our $\hat{\kappa}$ estimates conflate credit risk (κ) with the liquidity premium (ℓ) and any Islamic finance premium (π). In markets with thin secondary trading — particularly off-the-run corporate sukuk and sukuk from smaller GCC sovereigns — the liquidity component can dominate, making $\hat{\kappa}$ more a measure of illiquidity than of credit risk. A rigorous decomposition of $s = \kappa(1-\delta) + \ell + \pi$ would require panel data on bid-ask spreads, trading volumes, and on-the-run/off-the-run differentials that are not uniformly available for our sample. We note that for the primary benchmark κ-yield curves (GCC sovereigns, BNM sukuk), liquidity is high and this concern is attenuated.

Limitation 2: Recovery rate uncertainty. Our δ estimates are imposed from rating-agency recovery studies rather than estimated directly from sukuk default data. The empirical database of sukuk defaults is thin: the International Islamic Financial Market reports approximately [PLACEHOLDER: N_defaults] sukuk defaults over 2010–2025, insufficient for direct estimation of recovery rates. The robustness checks over $\delta \in \{0.20, 0.40, 0.60, 0.80\}$ bound the uncertainty, but a future study with a larger default database could allow δ to be treated as an unknown parameter jointly estimated with κ.

Limitation 3: Data vintage and market development. The pre-2015 period of our panel corresponds to an earlier stage of the sukuk market, when fewer issuers, shorter maturities, and thinner secondary markets characterised the landscape [**?**]. The NSS curves fitted to the early sub-period may be less precise than those for post-2018 data. A rolling-window analysis (not reported here, available on request) confirms that the main results are stronger in the 2015–2025 subsample, consistent with a maturing market.

7 Conclusion

We have constructed the first empirical κ-yield curve for Islamic fixed income markets, using a panel of [PLACEHOLDER: N] sukuk from GCC and Malaysian markets spanning 2010–2025. Our main findings are as follows.

(1) κ is identifiable. Point estimates $\hat{\kappa} = s/(1-\delta)$ are economically sensible, ranging from approximately [PLACEHOLDER: low] bps (AAA GCC sovereign at short maturities) to [PLACEHOLDER: high] bps (sub-investment grade corporate at long maturities) across

the credit spectrum. The estimates are stable over time, with the CIR model providing an adequate description of the mean-reverting dynamics (half-life of approximately [PLACEHOLDER: X] months for GCC sovereign κ).

(2) The κ-yield curve is well-behaved. It is typically upward-sloping for investment-grade issuers (in normal market conditions), inverts during stress periods (COVID-19, 2020), and exhibits the humped shape characteristic of mid-cycle credit conditions. Crucially, it is strictly positive throughout the sample — including during the European negative interest rate period — providing the riba-free property that conventional rate-anchored curves cannot deliver.

(3) κ-implied spreads outperform SOFR-based models. The κ-model achieves RMSE of [PLACEHOLDER: X] bps versus [PLACEHOLDER: Y] bps for the SOFR model (Diebold–Mariano $p < 0.01$). The outperformance is most pronounced during the negative-rate era and for long-maturity sukuk, the segments where interest-rate-anchored models are most problematic for Islamic finance practitioners.

(4) $\beta \approx 1$: empirical validation of ABIslam2026 Theorem 1. Panel regressions confirm that κ is the dominant driver of sukuk spreads ($\hat{\beta} \approx 1$, $H_0 : \beta = 1$ not rejected at the 5% level across all specifications), consistent with the theoretical prediction of ABIslam2026 Theorem 1. This is the first direct empirical validation of this result.

(5) CIR dynamics. κ is mean-reverting with half-life of approximately [PLACEHOLDER: X] months, confirming the stochastic κ extension proposed in ABIslam2026 Section 7.3. The Feller condition is satisfied for all countries, ensuring strict positivity of κ almost surely.

These findings have immediate implications for Islamic portfolio management. The κ-yield curve provides a complete, riba-free term structure of discount rates for valuing sukuk portfolios without any reference to interest-bearing benchmarks. Islamic fund managers can use κ-duration and κ-convexity (Equations (**??**)–(**??**)) as Shariah-compliant alternatives to modified duration and convexity. Central banks in OIC member countries can publish official κ-yield curves as riba-free benchmarks for their domestic sukuk markets, providing the term structure infrastructure that Islamic fixed income has lacked.

A natural extension of this work is to use the estimated κ values as calibration inputs to the iCDS pricing formula $s^* = \kappa(1 - \delta)$ (ABIslam2026 [**?**]), closing the loop between the empirical κ-yield curve and on-chain Islamic credit derivatives. A second extension is to estimate the κ-yield curve for corporate sukuk markets in non-GCC jurisdictions (Turkey, Bangladesh, Pakistan, sub-Saharan Africa), where the sukuk market is younger and the term structure of credit risk is less well documented. A third extension is to construct a κ-futures curve by applying the AHJ perpetual futures formula directly to traded Islamic derivatives,

providing a market-based measure of κ that complements the spread-based estimator of this paper.

Conflict of Interest

The first author (S. Ahmed) has developed an open-source smart-contract implementation of the κ-rate pricing framework described in this paper (Baraka Protocol, available at https://github.com/Arcus-Quant-Fund/BarakaDapp). No financial conflict of interest exists. The other authors declare no competing interests.

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